EVALUATING PINN-INSPIRED MONOTONICITY CONSTRAINTS FOR MULTI-TASK BEARING RUL PREDICTION ON XJTU-SY

Zishan Shaikh, Minal Jungare

Department of Artificial Intelligence and Data Science, Dr. D. Y. Patil Institute of Technology, Pimpri, Pune, India
Corresponding author: zishanshaikh20055002@gmail.com

Index Terms—remaining useful life, rolling bearings, multi-task learning, monotonicity constraints, XJTU-SY

Abstract

This study empirically compares five bidirectional long short-term memory (BiLSTM) configurations and training-only references for remaining useful life (RUL) prediction on XJTU-SY. The processed dataset has 15 bearing trajectories and 9,215 recordings; models use 32 vibration features and chronological 10-step windows. Historical M1, M2-original, and M4-original training loaders shuffled samples, whereas M3-original and M5-original preserved dataset order. Across three seeds, the unshuffled M2-control without monotonicity regularization obtained pooled mean absolute error/root mean squared error (MAE/RMSE) of 0.241/0.299, versus 0.251/0.313 for M3-original; a training-only unconstrained piecewise-linear age reference obtained 0.218/0.266. All neural pooled mean R-squared values were negative. The monotonicity penalty did not reduce adjacent-step or longer-lag upward violation rates for M3-original relative to M2-control. Label-free non-increasing isotonic projection can reduce saved-prediction errors but enforces shape by construction and is not a learned physical law. Paired whole-bearing bootstrap intervals over nine previously inspected test trajectories are exploratory. Larger penalty weights were not evaluated. These findings motivate cautious reporting of aggregation, temporal diagnostics, and training-order limitations.

I. INTRODUCTION

Machinery prognostics links data acquisition, health indicators, and remaining useful life (RUL) estimation [1]. Statistical data-driven estimation remains an established part of this field [2], while prognostics evaluation requires more than a single error score [3]. Data-driven RUL studies can report materially different conclusions depending on how repeated windows and bearing trajectories are aggregated. On small run-to-failure datasets, adjacent windows overlap, and a long trajectory can contribute many more test rows than a short one. Lower regression error also does not establish that predicted RUL decreases consistently over time.

We evaluate five fixed BiLSTM configurations under one leakage-controlled bearing split and compare them with a training-target mean and a training-only, unconstrained piecewise-linear age reference. We also report six exploratory matched-batching controls for M2-original and M4-original, using unshuffled training loaders. The test set had already been inspected; this analysis is a sensitivity check rather than an independent evaluation. M3-original remains strongest among the original neural configurations on pooled mean MAE/RMSE, but the unpenalized M2-original control trained without shuffling attains lower three-seed pooled means than historical M3-original. These observations do not identify a causal batch-order effect or establish an isolated monotonicity-penalty effect.

Bearing-RUL research has examined multi-task probabilistic estimation [4] and joint RUL prediction with fault diagnosis [5]. Physics-informed RUL methods have been reviewed [6], and recurrent models have been studied for long-sequence rolling-bearing prognosis [7]. Original physics-informed neural networks incorporate governing-equation information [8], while the wider physics-informed machine-learning literature covers several ways to use domain knowledge [9]. The present method uses a local temporal penalty only; these sources provide context, not a direct performance benchmark. No source paper's numerical result is presented as ours.

II. DATA AND METHODS

A. Dataset, Targets, and Features

The processed XJTU-SY benchmark [10] contains 15 bearing trajectories and 9,215 valid recordings across 35 Hz/12 kN, 37.5 Hz/11 kN, and 40 Hz/10 kN. Each record contains 32,768 samples from horizontal and vertical vibration channels. Files were numerically ordered. Bearing3_4 lacks index 1105; no file was fabricated, and subsequent source filenames were not renumbered. Existing valid files were assigned consecutive chronological positions.

For each trajectory with N valid records, the benchmark label is RUL_i = (N - 1 - i)/(N - 1), where i is the zero-based chronological position. A sequence of length 10 and stride 1 uses the final-step target RUL[t + 9]. Windows are created within bearings only. Each channel contributes mean, standard deviation, RMS, peak absolute amplitude, peak-to-peak amplitude, crest factor, kurtosis, skewness, dominant frequency, spectral centroid, spectral bandwidth, spectral RMS, and four band-energy ratios (0-1, 1-3, 3-6, and 6-12.8 kHz), for 16 features per channel and 32 total. Feature standardisation is fitted on the five training bearings and then applied to validation and test records.

B. Split and Model Configurations

Six bearings were candidates for leave-one-training-bearing-out fold selection: Bearing1_1, Bearing1_2, Bearing2_1, Bearing2_2, Bearing3_1, and Bearing3_2. In the fixed selected fold, the final fit uses Bearing1_1, Bearing1_2, Bearing2_1, Bearing3_1, and Bearing3_2 (5,764 sequences); validation uses the held-out candidate Bearing2_2 (152

sequences). Testing uses Bearing1_3, Bearing1_4, Bearing1_5, Bearing2_3, Bearing2_4, Bearing2_5, Bearing3_3, Bearing3_4, and Bearing3_5 (3,164 sequences). Each sequence contains 10 time steps and 32 features. Test bearings were excluded from scaler fitting, early stopping, and model selection. The project records no criterion for selecting Bearing2_2 from the leave-one-training-bearing-out folds; this fixed-fold choice is a limitation.

All models use a one-layer bidirectional LSTM [11] with 64 hidden units per direction. M1 predicts RUL only; M2-original adds a four-class fault head with fixed summed losses; M3-original adds a minibatch-local monotonicity penalty; M4-original uses learnable uncertainty-style task weights [12]; M5-original combines the M4-original weighting and M3-original penalty. The runs use Adam at 0.001, batch size 32, gradient clipping at 1.0, up to 40 epochs, patience 8, and seeds 42, 123, and 2024. RUL loss is mean squared error; the fault loss is cross-entropy. Fig. 1 summarises the shared pipeline, and Table 1 lists the objective differences.

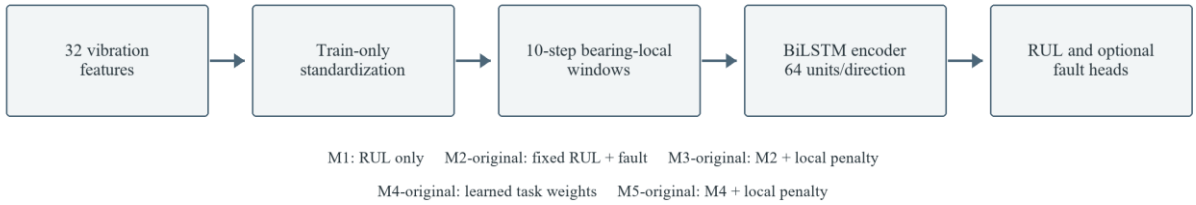


Fig. 1. Shared feature and sequence pipeline and the common BiLSTM encoder used by the evaluated configurations.

For M3-original and M5-original, the penalty is L_mono = mean[max(0, yhat_(b,j+1) - yhat_(b,j))] over adjacent available predictions from the same bearing within a minibatch. Predictions are grouped by bearing and sorted by sequence end position; transitions separated by a minibatch boundary are omitted. In the original three-seed runs, M1, M2-original and M4-original training loaders requested shuffling, whereas M3-original and M5-original preserved chronological dataset order. The six later controls changed the M2-original and M4-original training loader to unshuffled; validation and test loaders remained unshuffled. They were run in Colab on CPU (Python 3.13.16, PyTorch 2.11.0+cpu, NumPy 2.1.3, pandas 2.2.3, scikit-learn 1.6.1; CUDA unavailable). Thus original M2-original-to-M3-original and M4-original-to-M5-original comparisons vary both objectives and training-batch ordering. The title's term PINN-inspired denotes only a degradation-constraint motivation. No governing-equation, Joule's-law, or other physical residual is implemented; the XJTU-SY vibration inputs do not provide the current and resistance measurements needed to evaluate a Joule-heating residual.

Table 1. Evaluated objective definitions

Model	RUL objective	Fault objective	Additional term
M1	MSE	None	None
M2-original	MSE	Cross-entropy	Fixed sum
M3-original	MSE	Cross-entropy	0.1 L_mono
M4-original	Uncertainty weighted	Uncertainty weighted	Learned s_R, s_F
M5-original	Uncertainty weighted	Uncertainty weighted	0.1 L_mono

III. EXPERIMENTAL PROTOCOL

A. Reference Baselines and Metrics

The training-mean predictor is the arithmetic mean of training sequence-end RUL labels. The unconstrained piecewise-linear age predictor uses only observed sequence end position. It assigns equal total weight to each training bearing, fixes four knots at weighted training-position quantiles 0, 1/3, 2/3, and 1, and fits knot values by weighted least squares using training labels. Values outside training support are clamped to the nearest endpoint and predictions are clipped to [0, 1]. No validation or test labels were used to fit, tune, or select these references. The age fit has knot values 0.947, 0.345, 0.624, and 0.088, including an increasing middle segment; it is unconstrained and non-monotonic.

We report pooled window-weighted metrics, which give each sequence equal weight, and equal-bearing macro MAE/RMSE, which average nine per-bearing metrics equally. Neural results are three-seed means ± sample standard deviations; the baselines are deterministic single fits. R-squared is 1 - SSE/SST, with SST measured about the test-label mean. A negative value means squared error is worse than the diagnostic constant predictor equal to that test-label mean; because it requires test labels, that comparator is not deployable. We make no significance claims. This explicit use of error and trajectory-level summaries follows the broader concern that different prognostic metrics capture different properties [3].

IV. RESULTS

A. Pooled and Equal-Bearing RUL Results

Table 2 reports pooled window-weighted MAE, RMSE, and R-squared for the seven methods.

The unconstrained age reference has the best pooled MAE (0.218), RMSE (0.266), and R-squared (0.115). M3-original

is strongest among the five original neural configurations on pooled MAE/RMSE. In the matched-batching sensitivity analysis, M2-control has a mean pooled MAE of 0.241 and RMSE of 0.299 (Table 5); its MAE is slightly below the training mean (0.245), while its RMSE is higher than both simple references. Baseline/model rankings therefore depend on the metric and aggregation protocol.

The test contains 3,164 windows. Bearing3_4 supplies 1,505 windows, whereas Bearing2_4 supplies 33 windows. Therefore, pooled metrics weight trajectories very differently from the equal-bearing summary.

B. Operating-Condition Results

Table 4 and Fig. 3 show substantial condition variation. M3-original is lowest among original neural methods at 40 Hz/10 kN (0.266 ± 0.056), where M1, original M2-original, M4-original, and M5-original have higher mean MAE. Relative to original M2-original, M2-control has higher mean MAE at 35 Hz/12 kN and 37.5 Hz/11 kN but lower mean MAE at 40 Hz/10 kN (0.268 versus 0.391). Its 40 Hz/10 kN result resembles M3-original (0.266), but this does not establish a common underlying mechanism.

C. Matched-Batching Sensitivity Analysis

Table 5 reports pooled window-weighted and equal-bearing macro metrics. Delta denotes control minus comparator; negative MAE/RMSE deltas favor the control. M3-original and M5-original are historical unshuffled references. This is exploratory analysis on a previously inspected test set, without significance testing.

Across matched seeds, M2-control minus original M2-original changes mean pooled MAE by -0.056281 and RMSE by -0.082215; macro MAE/RMSE change by -0.005579/-0.004748. M4-control minus original M4-original changes mean pooled MAE/RMSE by -0.003422/-0.005480 and macro MAE/RMSE by +0.001879/+0.001422. For the three seed-matched unshuffled comparison, M2-control minus M3-original changes mean MAE/RMSE by -0.009906/-0.013606 and macro MAE/RMSE by -0.016902/-0.020487. For seeds 42/123/2024, the M2-control-minus-M3-original MAE/RMSE deltas are (+0.006236, +0.009197); (-0.030176, -0.041810); (-0.005779, -0.008205). Negative error deltas favor M2-control; these descriptive differences do not establish significance or a causal effect.

Condition-wise mean MAE also changes unevenly. M2-control is higher than original M2-original at 35 Hz/12 kN (0.215 vs 0.145) and 37.5 Hz/11 kN (0.191 vs 0.143), but lower at 40 Hz/10 kN (0.268 vs 0.391), resembling M3-original's 40 Hz/10 kN result (0.266). M4-control changes are smaller: 0.167 vs 0.128, 0.180 vs 0.177, and 0.376 vs 0.389 across those conditions. This pattern is descriptive and does not identify a mechanism.

D. Fault Classification and Temporal Diagnostics

Table 6 distinguishes the repeated sequence-row evaluation used for the neural macro-F1 from the bearing-level reference. Always predicting Outer gives sequence-row accuracy 0.195, macro precision 0.049, macro recall 0.250, and macro-F1 0.082; at the bearing level, accuracy is 0.444 and macro-F1 is 0.154 across nine bearings. Window rows repeat bearing labels and are not independent bearing examples. Labels denote the bearing's recorded final failure category, not fault onset. The selected training bearings contain Outer=3, Inner=1, Cage=0, and Mixed=1; absent Cage supervision is structural, and these results do not validate four-class diagnosis.

Table 7 reports within-bearing chronological adjacent comparisons. The training-mean predictor has 0/3,155 strict violations and zero adjacent change; both are degenerate consequences of a constant output, and its Spearman rho is undefined. The unconstrained age baseline has 1,345/3,155 strict violations, with mean positive increase 0.000748, mean bearing-wise rho -0.551, and equal-bearing mean absolute adjacent change 0.004006. These low-change statistics do not establish physical validity; the fitted age curve contains a non-monotonic segment.

M3-original has a strict violation rate of 0.490016 ± 0.008523, above M1 (0.478183 ± 0.003834), and does not reduce measured test-time violations. M4-original has the lowest mean rate among the five original neural configurations (0.473217 ± 0.009525). M2-control and M3-original have the same mean rate to six decimal places (0.490016), but their seed-level counts differ: M2-control has 1,541, 1,569, and 1,528 violations, whereas M3-original has 1,529, 1,577, and 1,532, each out of 3,155 transitions. Both count sets sum to 4,638, so their means coincide while their sample SDs differ. All neural/control adjacent-change entries in Table 7 use the equal-bearing aggregation defined in the caption; the alternative aggregate pooled over transitions is documented only in the metric audit.

For a lag L, an upward violation occurs when the prediction at an available endpoint L positions later is greater than the earlier prediction, assessed within each bearing. At lag 10, M2-control has macro and pooled violation rates of 0.371505 ± 0.028204 and 0.434396 ± 0.009525, while M3-original has 0.391790 ± 0.044763 and 0.447083 ± 0.021156, respectively, across 3,074 pooled transitions. At lag 20, the corresponding macro rates are 0.320773 ± 0.040856 and 0.354928 ± 0.061644, and pooled rates are 0.405608 ± 0.023296 and 0.431189 ± 0.033027, across 2,984 transitions. All nine test bearings are eligible; Bearing2_4 contributes 23 and 13 transitions at lags 10 and 20 because it has 33 windows. Longer-lag violations are less frequent than adjacent-step violations but remain substantial. M3-original does not reduce these rates relative to M2-control. These descriptive diagnostics use a previously inspected test set and do not identify a causal loss effect because historical objective and training-order differences remain relevant.

Table 2. Pooled window-weighted test results

Method	MAE	RMSE	R-squared
Training mean	0.245	0.283	-0.001
Piecewise-linear age	0.218	0.266	0.115
M1	0.310 ± 0.005	0.400 ± 0.004	-1.000 ± 0.042
M2-original	0.298 ± 0.014	0.381 ± 0.011	-0.813 ± 0.110
M3-original	0.251 ± 0.023	0.313 ± 0.035	-0.229 ± 0.266
M4-original	0.305 ± 0.013	0.385 ± 0.014	-0.853 ± 0.137
M5-original	0.290 ± 0.014	0.361 ± 0.014	-0.623 ± 0.127

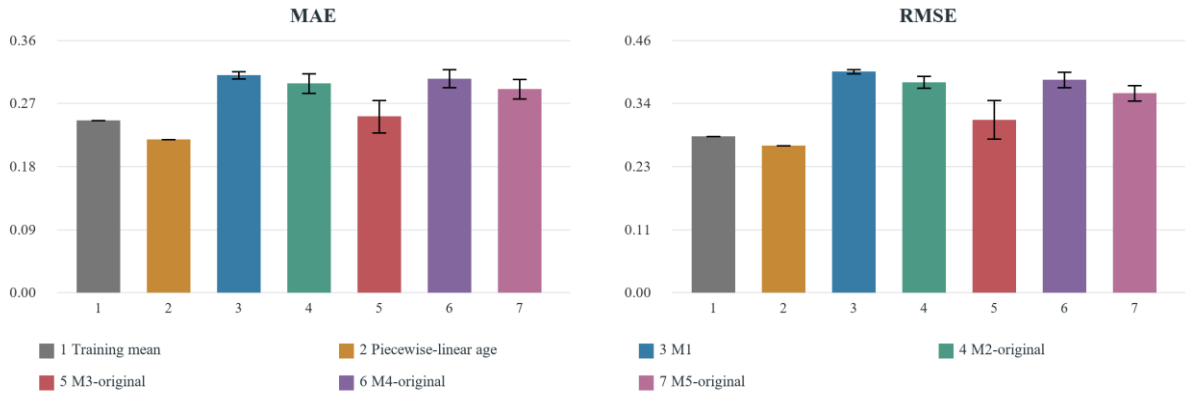


Fig. 2. Pooled test MAE and RMSE; baseline fits are deterministic, neural error bars show sample standard deviation across seeds.

Table 3. Equal-bearing macro test errors

Method	Macro MAE	Macro RMSE
Training mean	0.236	0.274
Piecewise-linear age	0.238	0.275
M1	0.239 ± 0.007	0.290 ± 0.007
M2-original	0.230 ± 0.007	0.278 ± 0.010
M3-original	0.241 ± 0.023	0.294 ± 0.027
M4-original	0.237 ± 0.009	0.288 ± 0.010
M5-original	0.242 ± 0.019	0.292 ± 0.022

Table 4. Test MAE by operating condition

Condition	Method	MAE
35 Hz / 12 kN	Training mean	0.234
35 Hz / 12 kN	Piecewise-linear age	0.141
35 Hz / 12 kN	M1	0.147 ± 0.015
35 Hz / 12 kN	M2-original	0.145 ± 0.017
35 Hz / 12 kN	M3-original	0.245 ± 0.064
35 Hz / 12 kN	M4-original	0.128 ± 0.011
35 Hz / 12 kN	M5-original	0.201 ± 0.044
37.5 Hz / 11 kN	Training mean	0.244
37.5 Hz / 11 kN	Piecewise-linear age	0.265
37.5 Hz / 11 kN	M1	0.143 ± 0.003
37.5 Hz / 11 kN	M2-original	0.143 ± 0.037
37.5 Hz / 11 kN	M3-original	0.221 ± 0.060
37.5 Hz / 11 kN	M4-original	0.177 ± 0.027
37.5 Hz / 11 kN	M5-original	0.188 ± 0.022
40 Hz / 10 kN	Training mean	0.247
40 Hz / 10 kN	Piecewise-linear age	0.209
40 Hz / 10 kN	M1	0.411 ± 0.007
40 Hz / 10 kN	M2-original	0.391 ± 0.010
40 Hz / 10 kN	M3-original	0.266 ± 0.056
40 Hz / 10 kN	M4-original	0.389 ± 0.011
40 Hz / 10 kN	M5-original	0.349 ± 0.012

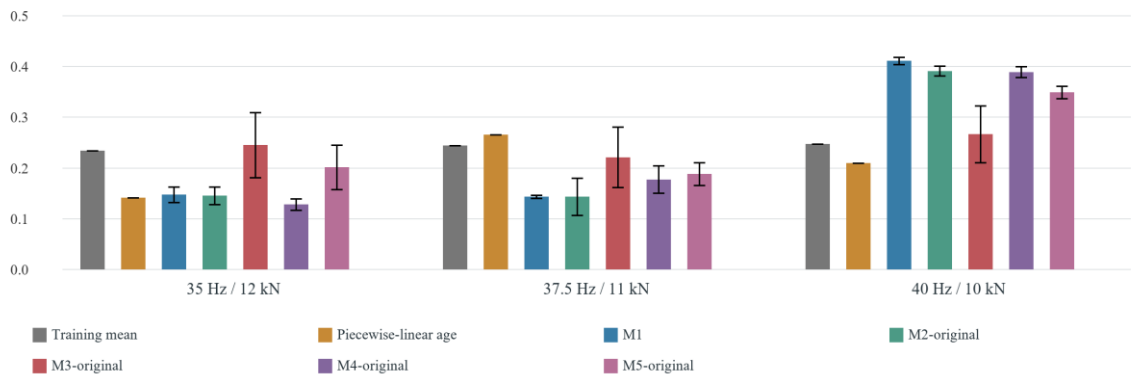


Fig. 3. Test MAE by operating condition for the two deterministic references and five neural configurations.

Table 5. Matched-batching sensitivity results (mean ± sample SD, three seeds); S/U denote train shuffle=True/False.

Run condition	Pooled MAE	Pooled RMSE	Pooled R-squared	Macro MAE	Macro RMSE
M2-original (S)	0.298 ± 0.014	0.381 ± 0.011	-0.813 ± 0.110	0.230 ± 0.007	0.278 ± 0.010
M2-control (U)	0.241 ± 0.032	0.299 ± 0.047	-0.133 ± 0.365	0.224 ± 0.006	0.274 ± 0.006
M3-original (U)	0.251 ± 0.023	0.313 ± 0.035	-0.229 ± 0.266	0.241 ± 0.023	0.294 ± 0.027
M4-original (S)	0.305 ± 0.013	0.385 ± 0.014	-0.853 ± 0.137	0.237 ± 0.009	0.288 ± 0.010
M4-control (U)	0.301 ± 0.021	0.380 ± 0.027	-0.805 ± 0.255	0.239 ± 0.015	0.290 ± 0.014
M5-original (U)	0.290 ± 0.014	0.361 ± 0.014	-0.623 ± 0.127	0.242 ± 0.019	0.292 ± 0.022

Table 6. Exploratory fault-class reference and neural macro-F1

Method	Evaluation unit	Macro-F1
Always Outer (training majority)	Sequence rows (n=3,164)	0.082
Always Outer (training majority)	Bearing level (n=9)	0.154
M1	No fault head	N/A — no fault head.
M2-original	Sequence rows (n=3,164/seed)	0.084 ± 0.025
M3-original	Sequence rows (n=3,164/seed)	0.251 ± 0.170
M4-original	Sequence rows (n=3,164/seed)	0.105 ± 0.038
M5-original	Sequence rows (n=3,164/seed)	0.071 ± 0.002

Table 7. Test-time temporal diagnostics. Neural/control means ± sample SD; adjacent change is the equal-bearing mean of within-bearing mean absolute changes.

Method	Violation rate	Mean positive violation	Signed Spearman rho	Mean |adjacent change|
Training mean	0.0000	0.000000	N/A	0.000000
Piecewise-linear age	0.4263	0.000748	-0.551	0.004006
M1	0.478183 ± 0.003834	0.007140 ± 0.000308	-0.653885 ± 0.034477	0.01537 ± 0.00026
M2-original	0.485261 ± 0.006820	0.013410 ± 0.001105	-0.647080 ± 0.051921	0.02588 ± 0.00075
M3-original	0.490016 ± 0.008523	0.015791 ± 0.002894	-0.547280 ± 0.150756	0.02164 ± 0.00424
M4-original	0.473217 ± 0.009525	0.011943 ± 0.002341	-0.667670 ± 0.025871	0.02241 ± 0.00192
M5-original	0.493608 ± 0.004619	0.018917 ± 0.002735	-0.679674 ± 0.011775	0.03130 ± 0.00409
M2-control	0.490016 ± 0.006641	0.014036 ± 0.001167	-0.602180 ± 0.103743	0.02010 ± 0.00399
M4-control	0.490650 ± 0.007262	0.019408 ± 0.001875	-0.647338 ± 0.066543	0.03379 ± 0.00542

Table 8. Exploratory paired whole-bearing bootstrap: mean across 1,000 bootstrap replicates [95% percentile interval].

Method	Variant	Macro MAE	Macro RMSE	Pooled MAE	Pooled RMSE
Piecewise-linear age	Raw	0.239 [0.166, 0.312]	0.276 [0.206, 0.348]	0.219 [0.187, 0.261]	0.268 [0.239, 0.309]
Piecewise-linear age	Isotonic	0.224 [0.152, 0.302]	0.255 [0.186, 0.332]	0.196 [0.167, 0.215]	0.233 [0.210, 0.257]
M2-control	Raw	0.223 [0.176, 0.279]	0.273 [0.218, 0.335]	0.239 [0.194, 0.264]	0.295 [0.246, 0.327]
M2-control	Isotonic	0.219 [0.173, 0.274]	0.265 [0.212, 0.326]	0.234 [0.190, 0.261]	0.289 [0.238, 0.321]
M3-original	Raw	0.240 [0.195, 0.290]	0.293 [0.240, 0.353]	0.251 [0.223, 0.275]	0.311 [0.276, 0.345]
M3-original	Isotonic	0.234 [0.190, 0.284]	0.283 [0.231, 0.340]	0.245 [0.217, 0.266]	0.302 [0.267, 0.331]

E. Isotonic Projection and Bearing-Bootstrap Uncertainty

A non-increasing isotonic projection was fitted independently to each complete bearing trajectory using only chronological endpoint positions and saved predictions; true RUL labels were not used to fit the projection. The previously analysed M1 file reproduced the expected pooled-MAE change from 0.306988 to 0.298858 before batch processing.

Projection lowers the saved-data point estimates for all three displayed methods, but it enforces shape by construction and is not evidence of a physical degradation law. Neural bootstrap metrics were calculated per seed and then averaged; identical bearing resamples were used across methods. With nine test bearings and prior inspection of this test set, the intervals describe exploratory sampling sensitivity rather than confirmatory uncertainty.

V. DISCUSSION AND LIMITATIONS

Simple references remain essential to interpretation. The RUL construction assigns each bearing a linear decline from first to final valid record, so chronological position is informative. However, the fitted piecewise-linear age

predictor is unconstrained and has a non-monotonic segment. The training mean has a pooled MAE slightly above M2-control but lower pooled RMSE; the age reference remains best on pooled MAE/RMSE. Equal-bearing aggregation changes rankings, underscoring that overlapping windows are not independent bearing trajectories.

The original M2-original-to-M3-original and M4-original-to-M5-original comparisons confound objective with order: M1, M2-original and M4-original shuffle; M3-original and M5-original preserve dataset order. Controls remain exploratory, with three seeds, a previously inspected test set, no significance tests, and unverified hardware parity because original-run device metadata was not retained. Only lambda=0.1 was evaluated as a nonzero monotonicity-penalty weight; sensitivity to larger penalty weights remains untested and is left for future work. Other limits are 15 trajectories, one fixed validation fold with no recorded selection rationale, few independent bearings, no Cage examples in training, negative mean neural pooled R-squared, condition-dependent errors, and bearing-level labels without fault-onset annotations. The controls do not establish causal batch-order or penalty effects.

VI. CONCLUSION

This fixed-split empirical comparison finds M3-original strongest among the five original neural configurations on pooled mean MAE/RMSE, while the unconstrained age reference remains best on pooled error and the training-mean reference remains competitive. M2-control shows that the original neural ranking is sensitive to training-batch ordering, especially for pooled errors, but does not establish a causal effect. Results vary by aggregation and operating condition; all neural pooled mean R-squared values remain negative.

The monotonicity penalty does not reduce measured strict test-time violations in these results, including at lags 10 and 20 relative to M2-control. These previously inspected test-set diagnostics remain exploratory and cannot isolate a causal loss effect. Broader bearing-level validation and Cage-supervised data are needed before claims about generalization or four-class diagnosis. Future work will investigate physics residuals based on characteristic bearing defect frequencies (BPFO/BPFI) and a calibrated exponential-degradation ordinary differential equation, subject to the availability of suitable physical parameters and independent validation.

DATA AND CODE AVAILABILITY

The processed data, experiment configurations, and saved results used here are retained in the associated research project. Run manifests and SHA-256 artifact hashes are retained for integrity checks. The raw XJTU-SY collection is identified in [10]. No public code repository or DOI is stated in the project records.

FUNDING AND COMPETING INTERESTS

Funding: This research received no external funding. Competing interests: The author declares no competing interests.

REFERENCES

[1] Y. Lei, N. Li, L. Guo, N. Li, T. Yan, and J. Lin, 'Machinery health prognostics: A systematic review from data acquisition to RUL prediction,' Mechanical Systems and Signal Processing, vol. 104, pp. 799-834, 2018, doi: 10.1016/j.ymssp.2017.11.016.

[2] X.-S. Si, W. Wang, C.-H. Hu, and D.-H. Zhou, 'Remaining useful life estimation - A review on the statistical data driven approaches,' European Journal of Operational Research, vol. 213, no. 1, pp. 1-14, 2011, doi: 10.1016/j.ejor.2010.11.018.

[3] M. L. Baptista, S. Panse, and B. F. Santos, 'Revision and implementation of metrics to evaluate the performance of prognostics models,' Measurement, vol. 236, Art. no. 115038, 2024, doi: 10.1016/j.measurement.2024.115038.

[4] X. Wang, Y. Li, K. Noman, and A. K. Nandi, 'Multi-task learning mixture density network for interval estimation of the remaining useful life of rolling element bearings,' Reliability Engineering and System Safety, vol. 251, Art. no. 110348, 2024, doi: 10.1016/j.ress.2024.110348.

[5] L. Zhou and H. Wang, 'Multi-task model of adaptive multi-scale feature fusion and adaptive mixture-of-experts for equipment remaining useful life prediction and fault diagnosis,' Expert Systems with Applications, vol. 272, Art. no. 126807, 2025, doi: 10.1016/j.eswa.2025.126807.

[6] H. Li, Z. Zhang, T. Li, and X. Si, 'A review on physics-informed data-driven remaining useful life prediction: Challenges and opportunities,' Mechanical Systems and Signal Processing, vol. 209, Art. no. 111120, 2024, doi: 10.1016/j.ymssp.2024.111120.

[7] Q. Zhao, X. Zhang, X. Luo, W. Liang, and E. Mbeka, 'A physics-informed recurrent neural network for long sequence remaining useful life prediction of rolling bearing with implicit function,' Reliability Engineering and System Safety, vol. 274, Art. no. 112412, 2026, doi: 10.1016/j.ress.2026.112412.

[8] M. Raissi, P. Perdikaris, and G. E. Karniadakis, 'Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations,' Journal of Computational Physics, vol. 378, pp. 686-707, 2019, doi: 10.1016/j.jcp.2018.10.045.

[9] G. E. Karniadakis, I. G. Kevrekidis, L. Lu, P. Perdikaris, S. Wang, and L. Yang, 'Physics-informed machine learning,' Nature Reviews Physics, vol. 3, no. 6, pp. 422-440, 2021, doi: 10.1038/s42254-021-00314-5.

[10] B. Wang, Y. Lei, N. Li, and N. Li, 'A hybrid prognostics approach for estimating remaining useful life of rolling element bearings,' IEEE Transactions on Reliability, vol. 69, no. 1, pp. 401-412, 2020, doi: 10.1109/TR.2018.2882682.

[11] S. Hochreiter and J. Schmidhuber, 'Long short-term memory,' Neural Computation, vol. 9, no. 8, pp. 1735-1780, 1997, doi: 10.1162/neco.1997.9.8.1735.

[12] A. Kendall, Y. Gal, and R. Cipolla, 'Multi-task learning using uncertainty to weigh losses for scene geometry and semantics,' in Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition, 2018, pp. 7482-7491, doi: 10.1109/CVPR.2018.00781.